



**A PROJECT REPORT ON**  
**“YouTube Trending Analytics “**

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## 1. Introduction

This project analyzes YouTube Trending videos across multiple countries to identify patterns in engagement, category performance, regional popularity, and sentiment. It includes data cleaning using Python, SQL analytics, and visual storytelling with Power BI.

## 2. Dataset Overview

The dataset contains trending video metadata, including title, channel, views, likes, comments, publish time, category, description, and regional information. Category JSON files were used to map category IDs to meaningful labels.

## 3. Data Cleaning (Python)

- Removed duplicates and invalid rows
- Standardized date formats
- Cleaned text fields
- Added title\_length, tag\_count, sentiment\_label
- Merged multi-country datasets
- Exported cleaned\_sql\_ready.csv

## **4. SQL Analytics**

A MySQL database was created to store cleaned data. SQL queries were used to extract insights such as average views by category, top-performing channels, sentiment distribution, trending duration, and time-series view patterns. CSV outputs were used in Power BI.

## **5. Power BI Dashboard**

The dashboard includes KPI cards, trending views line chart, sentiment donut chart, likes vs views scatter, category analysis bar charts, top channels, geographic map visual, and engagement metrics—providing a complete analytics view.

## **6. Key Insights**

- Positive sentiment videos receive higher engagement.
- Music and Entertainment categories dominate.
- India and the US lead in overall view count.
- Top channels contribute the highest viewer retention.
- Strong correlation found between likes and views.
- Clear regional patterns observed using time-series and map visuals.

## **7. Conclusion**

This project demonstrates an end-to-end analytics pipeline including data cleaning, SQL analysis, sentiment processing, and dashboard visualization. It effectively highlights YouTube viewership trends and provides actionable insights for content performance evaluation.

## 8.Tools Used

Python, MySQL, Power BI, CSV/JSON datasets, Google Colab.

## 9.Project Structure

YouTube\_Trending\_Analytics/

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